

PredictFun: A No-Subsidy Prediction Market Launchpad

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Abstract

This paper specifies a prediction market launchpad in which binary markets form through a pre-graduation Logarithmic Market Scoring Rule (LMSR) pricing curve without protocol subsidy or guaranteed fill. Markets can form real betting activity immediately, but pre-graduation bets are provisional receipts rather than final outcome-token positions. Only cleared exposure becomes a Conditional Token Framework (CTF)-style outcome-token claim.

A market graduates only when enough path-compatible opposing demand exists to form a fully collateralized settlement pool. At graduation, deterministic path-overlap clearing splits receipts into LMSR path segments, matches YES and NO segments that traversed the same probability bands in opposite directions, and refunds incompatible or excess segments. Matched exposure converts into fully backed outcome tokens while each receipt's retained and refunded amounts preserve the actual LMSR cost basis paid by the bettor.

The design combines the continuous price discovery of LMSR with path-overlap batch clearing. LMSR gives every market a smooth probability curve before graduation. Batch clearing prevents insolvency by ensuring that final redeemable claims are backed by real collateral rather than virtual liquidity. This paper focuses on the launch, receipt, clearing, and solvency mechanism. Post-graduation venue design and final oracle governance are modular protocol layers.

1. Introduction

Prediction market launchpads inherit a tension from two different market types.

Token launchpads use bonding curves to create immediate trading. The curve can quote a token price before deep two-sided liquidity exists because the token has no fixed redemption value.

Prediction markets are different. A YES share or NO share is a contingent claim:

YES $\mapsto$ 1	if the event resolves YES
NO $\mapsto$ 1	if the event resolves NO

For a binary fixed-payout market, a collateral-complete pair satisfies:

$$1 \text{ YES} + 1 \text{ NO} = 1 \text{ unit of collateral}$$

If a mechanism sells 100 YES at a price of 0.30, it collects 30 units of collateral. If YES wins, those shares owe 100 units. Without a subsidy, liquidity provider inventory, or compatible NO demand, the missing 70 units are an insolvency.

This paper proposes a no-subsidy alternative:

Before graduation: LMSR-priced provisional receipts.

At graduation: path-overlap batch clearing and refunds.

After graduation: retained CTF-style outcome tokens enter a standard prediction market venue.

The mechanism does not promise instant fill. It promises instant participation in price discovery, then deterministic clearing once the market has enough real opposing demand. If that demand never appears, the protocol refunds receipts rather than pretending that an underfunded market can settle as a fixed-payout prediction market.

2. Design Goals

The mechanism is designed around seven goals:

1. **No protocol subsidy.** The launchpad should not need to bankroll the missing side of early prediction bets.
2. **Continuous pre-grad pricing.** Bettors should see a live probability curve rather than a static pool.
3. **Explicit provisionality.** Pre-graduation participants should know they hold refund-backed receipts, not final outcome tokens.
4. **Deterministic lifecycle.** Every market should have a defined path for graduation, non-graduation, cancellation, and refund.
5. **No hidden insolvency.** A final fixed-payout claim should never exist unless it can be fully backed.
6. **Cost-basis preserving partial fills.** If a provisional receipt partially survives graduation, retained and refunded path segments should add back to the actual LMSR cost basis paid by the bettor.
7. **Resolution independence.** Trading mechanics and truth resolution should be separate layers.

The design intentionally omits trading fees. Fees can be added later, but introducing them before the fill/refund mechanism is settled makes user accounting harder to reason about.

3. Background: Logarithmic Market Scoring Rule, Conditional Token Framework Outcome Tokens, And Pari-Mutuel Markets

Logarithmic Market Scoring Rule

The Logarithmic Market Scoring Rule (LMSR) gives the pre-graduation market a smooth probability curve whose marginal prices move continuously with YES and NO demand.

For a binary market:

$$C(q_{\text{yes}}, q_{\text{no}}) = b \ln \left(e^{q_{\text{yes}}/b} + e^{q_{\text{no}}/b} \right)$$

The marginal YES price is:

$$P_{\text{yes}} = \frac{e^{q_{\text{yes}}/b}}{e^{q_{\text{yes}}/b} + e^{q_{\text{no}}/b}}$$

This follows by differentiating the cost function:

$$\begin{aligned} \frac{dC}{dq_{\text{yes}}} &= b \left(\frac{1}{e^{q_{\text{yes}}/b} + e^{q_{\text{no}}/b}} \right) \left(\frac{1}{b} \right) e^{q_{\text{yes}}/b} \\ &= \frac{e^{q_{\text{yes}}/b}}{e^{q_{\text{yes}}/b} + e^{q_{\text{no}}/b}} \end{aligned}$$

The marginal NO price is:

$$P_{\text{no}} = 1 - P_{\text{yes}}$$

The cost of buying additional provisional shares is:

$$\text{cost} = C(q_{\text{after}}) - C(q_{\text{before}})$$

Equivalently, the trade cost is the integral of the marginal LMSR price over the trade path:

$$\begin{aligned} \text{cost}_{\text{yes}}(s) &= \int_0^s P_{\text{yes}}(q_{\text{yes}} + x, q_{\text{no}}) dx \\ &= C(q_{\text{yes}} + s, q_{\text{no}}) - C(q_{\text{yes}}, q_{\text{no}}) \end{aligned}$$

In a conventional LMSR automated market maker, the liquidity parameter b implies a bounded market-maker loss. That model requires a funded subsidy. In this design, b has a different role: it controls the smoothness of pre-graduation pricing and how strongly early demand moves the displayed probability.

Conditional Token Framework Outcome Tokens

In a Conditional Token Framework (CTF)-style prediction market, the payoff of a retained position is known:

$$\begin{aligned} 1 \text{ retained YES} &\mapsto 1 && \text{if YES resolves true} \\ 1 \text{ retained NO} &\mapsto 1 && \text{if NO resolves true} \end{aligned}$$

This is the market structure users expect from Polymarket-style YES/NO trading. The user does not merely own a pro-rata claim on a pot. They own a side-specific outcome token with a fixed terminal payout.

For a no-subsidy launchpad, this creates a constraint:

retained YES exposure and retained NO exposure must be jointly collateral-complete.

Pari-Mutuel Contrast

In a pure pari-mutuel market, winners share the pool and final odds are not fully known until betting closes. That is solvent, but it is not the same as a fixed-payout YES/NO market.

PredictFun uses a related solvency discipline but not the same terminal payoff rule. Excess demand is refunded before settlement. The remaining demand converts into CTF-style fixed-payout outcome tokens.

The hybrid can be summarized as:

LMSR for pre-grad price discovery.
Path-overlap clearing for CTF-style outcome-token fill.
Optimistic oracle for truth.

4. Mechanism Overview

Each market has four protocol stages:

Draft and review.
Pre-graduation receipt formation.
Graduation clearing.
Post-clearing settlement or refund.

Pre-Graduation

Bettors buy provisional YES or NO receipts against an LMSR curve. Each receipt records:

side
provisional shares
cost basis
average entry price for display
LMSR state before the receipt
LMSR state after the receipt
path interval traversed by the receipt

A receipt is not a final redeemable position. It is a claim to participate in graduation clearing.

The displayed LMSR quote remains useful:

price $\approx$ implied probability

But the quote is not a fill guarantee. It is the price path used to decide which receipt segments can be retained and how refunds are computed.

Graduation

The market graduates when path-compatible YES/NO demand reaches a threshold. Compatibility means opposing receipts traversed the same LMSR path band in opposite directions. Over such a band, the YES marginal price and NO marginal price are complements:

$$P_{\text{yes}}(r) + P_{\text{no}}(r) = 1$$

At graduation:

1. New pre-grad betting stops.
2. The frozen receipt book is split into LMSR path bands.
3. YES and NO segments active over the same band are matched.
4. Crowded-side segments are retained pro-rata within that band.
5. Unmatched and overbet segments are refunded at their actual path cost.
6. Retained exposure becomes CTF-style YES/NO outcome tokens.
7. The market hands backed tokens to a standard post-graduation venue.

5. Protocol Lifecycle

The protocol lifecycle must define what happens not only when a market succeeds, but also when it fails to attract enough path-compatible demand. The v0.2 design uses the following states:

State	Entry Condition	Exit Condition	User Result
Draft	Creator proposes a market.	Market review approves, rejects, or requests edits.	No trading exposure exists.
Approved	Canonical market rules and source hierarchy are accepted.	Creator posts required bond and market opens.	No receipt exposure exists yet.
PreGradOpen	Market accepts pre-graduation receipt purchases.	Graduation eligibility is reached, the pre-grad close time arrives, or the market is canceled.	Users hold provisional receipts backed by escrowed cost basis.
FrozenForClearing	Anyone freezes an eligible market for deterministic clearing.	Clearing algorithm completes.	No new receipts can enter; all existing lots are final clearing inputs.
Graduated	Cleared filled market cap reaches the graduation threshold.	Backed outcome tokens are delivered to retained bettors and refunds are paid.	Retained exposure becomes fixed-payout YES/NO claims.
NotGraduated	The pre-grad close time arrives without enough path-compatible demand.	Refund batch completes.	All active receipts are refunded at cost basis, excluding any explicitly disclosed fees.
Canceled	Market is rejected after approval, source rules fail before clearing, or emergency cancellation is triggered.	Refund batch completes and any creator-bond penalty is applied.	Active receipts are refunded; no outcome exposure exists.

The key lifecycle rule is:

No receipt can become a resolution-bearing position except through graduation clearing.

This keeps the protocol from accidentally creating two different products: provisional price discovery before graduation and fixed-payout prediction markets after graduation.

Market Deadlines

Each market should define:

```

pregrad_open_time
pregrad_close_time
event_end_time
resolution_deadline

```

The pre-graduation close time must be before the market can be resolved and should usually be before the event outcome is knowable. The protocol should not allow a market to graduate after the relevant outcome is already public.

Graduation Eligibility

A market becomes eligible for clearing when the current receipt book can produce enough path-compatible filled market cap:

$$F_candidate \geq graduation_threshold$$

and any non-price gates are satisfied:

```
rule_quality_gate = true
time_remaining_gate = true
concentration_gate = true
market_not_canceled = true
```

Once eligibility is met, anyone can freeze the market for clearing. Freezing is important because clearing must run over a fixed set of receipt lots. New receipts cannot enter after the freeze.

Non-Graduation

If the pre-grad close time arrives before the market becomes eligible, the market does not produce resolution-bearing outcome tokens. The protocol refunds all active receipts:

```
retained_shares = 0
refund = cost_basis
```

This is the cleanest no-subsidy rule. A future protocol version could support a micro-clearing path for small matched books below the graduation threshold, but v0.2 treats the graduation threshold as the minimum viable fixed-payout market size.

6. Pre-Graduation Receipt Rights And Accounting

A pre-graduation receipt is the accounting object created when a bettor moves the LMSR state. It is not a CTF token yet and it does not by itself represent a redeemable claim. It records the bettor's priced intent so graduation clearing can later decide how much of that intent is backed by path-compatible opposing demand.

In v0.2, a receipt gives the holder three rights:

1. A right to participate in deterministic graduation clearing.
2. A right to receive retained outcome tokens if the receipt is matched.
3. A right to refund for any unretained cost basis.

A receipt does not give the holder:

- A guaranteed fill.
- A guaranteed final YES or NO position.
- A right to resolution payout before graduation.
- A claim on creator bonds, resolver bonds, or protocol insurance.
- A right to force the market to graduate.

To keep the mechanism simple, v0.2 receipts are non-transferable and cannot be voluntarily withdrawn before clearing or cancellation. Transferability, secondary receipt markets, or pre-

grad exits can be added later, but they require separate rules for lot history, price manipulation, and refund ownership.

The receipt variables come from three places:

Variable	Source	Meaning
b	Market parameter	Controls LMSR curve smoothness and price sensitivity.
q_yes, q_no	LMSR state	Running score state used to quote marginal prices.
s_l	Trade size	Provisional shares bought in receipt lot l.
c_l	LMSR cost difference	Collateral paid for receipt lot l.
p_l	Derived from receipt	Average entry price for display, equal to c_l / s_l .
q_before_l	Receipt state	LMSR state before lot l was created.
q_after_l	Receipt state	LMSR state after lot l was created.
r_l	Derived path coordinate	$q_{yes} - q_{no}$ over the lot's traversed path.
Y, N	Receipt aggregates	Total provisional YES and NO shares.

For each receipt lot l, the market tracks:

owner_l = receipt holder
 side_l in {YES, NO}
 s_l = provisional shares
 c_l = cost basis
 $p_l = c_l / s_l$, for display and user accounting
 q_yes_before_l, q_no_before_l
 q_yes_after_l, q_no_after_l
 $r_{low_l} = \min(q_{yes_before_l} - q_{no_before_l}, q_{yes_after_l} - q_{no_after_l})$
 $r_{high_l} = \max(q_{yes_before_l} - q_{no_before_l}, q_{yes_after_l} - q_{no_after_l})$
 sequence_l = deterministic lot ordering key
 status_l in {active, retained, refunded}

The cost basis is escrowed when the receipt is created:

escrowed_collateral_l = c_l

Escrowed receipt collateral cannot be used as creator bond, resolver bond, insurance capital, or protocol working capital. It has only two destinations:

refund to receipt holder
 locked collateral backing retained outcome tokens

The aggregate quantities are:

Y = sum YES shares
 N = sum NO shares
 C_y = sum YES cost basis
 C_n = sum NO cost basis
 C_total = C_y + C_n

The displayed LMSR state can use Y and N directly:

$$\begin{aligned} q_{\text{yes}} &= Y \\ q_{\text{no}} &= N \end{aligned}$$

More generally, a market can include an initial pricing offset:

$$\begin{aligned} q_{\text{yes}} &= q_{\text{yes},0} + Y \\ q_{\text{no}} &= q_{\text{no},0} + N \end{aligned}$$

For a neutral market, $q_{\text{yes},0} = q_{\text{no},0} = 0$, so the initial price is 0.50. For a market that should open at initial probability P_0 , only the difference between the offsets matters:

$$\begin{aligned} P_0 &= \frac{e^{q_{\text{yes},0}/b}}{e^{q_{\text{yes},0}/b} + e^{q_{\text{no},0}/b}} \\ q_{\text{yes},0} - q_{\text{no},0} &= b \ln \left(\frac{P_0}{1 - P_0} \right) \end{aligned}$$

The offset is still just LMSR state. It is not a subsidy and does not create backed exposure.

Given a current state $(q_{\text{yes}}, q_{\text{no}})$, a YES receipt for s_l provisional shares is priced as:

$$\begin{aligned} q_{\text{before}} &= (q_{\text{yes}}, q_{\text{no}}) \\ q_{\text{after}} &= (q_{\text{yes}} + s_l, q_{\text{no}}) \\ c_l &= C(q_{\text{after}}) - C(q_{\text{before}}) \\ p_l &= \frac{c_l}{s_l} \end{aligned}$$

A NO receipt is symmetric:

$$\begin{aligned} q_{\text{before}} &= (q_{\text{yes}}, q_{\text{no}}) \\ q_{\text{after}} &= (q_{\text{yes}}, q_{\text{no}} + s_l) \\ c_l &= C(q_{\text{after}}) - C(q_{\text{before}}) \\ p_l &= \frac{c_l}{s_l} \end{aligned}$$

So p_l is the average LMSR price paid across the trade, not necessarily the final marginal quote after the trade. This matters because graduation clearing cannot treat the whole receipt as a uniform-price block. The receipt's cost basis is distributed along the LMSR path it traversed.

If the user enters by budget instead of share amount, the share amount can be solved from the same LMSR cost equation. For a YES budget c , define:

$$\begin{aligned} A &= e^{q_{\text{yes}}/b} \\ B &= e^{q_{\text{no}}/b} \end{aligned}$$

Then:

$$\begin{aligned}
c &= b \ln \left(\frac{Ae^{s/b} + B}{A + B} \right) \\
e^{c/b}(A + B) &= Ae^{s/b} + B \\
e^{s/b} &= \frac{e^{c/b}(A + B) - B}{A} \\
s &= b \ln \left(\frac{e^{c/b}(A + B) - B}{A} \right)
\end{aligned}$$

For a NO budget, swap A and B:

$$s = b \ln \left(\frac{e^{c/b}(A + B) - A}{B} \right)$$

Receipts can be split by their path endpoints. This gives the clearing algorithm a deterministic finite set of bands even though LMSR pricing is continuous. Within each band, the algorithm knows which YES receipts and which NO receipts were active over that same segment of the curve.

7. Path Compatibility And Filled Market Cap

A global overbet scale is not enough for fixed-payout clearing. If one side bought across many different LMSR prices, the protocol cannot safely lump those receipts into one average odds pool and give the crowded side a simple pro-rata haircut. That can retain exposure from curve regions that were never funded by path-compatible opposite-side demand.

The clearing primitive is the LMSR path segment. Define:

$$\begin{aligned}
r &= q_{\text{yes}} - q_{\text{no}} \\
P_{\text{yes}}(r) &= 1 / (1 + \exp(-r / b)) \\
P_{\text{no}}(r) &= 1 - P_{\text{yes}}(r)
\end{aligned}$$

A YES buy increases r . A NO buy decreases r . A receipt for s_l provisional shares therefore traverses an interval:

$$\begin{aligned}
&[r_{\text{low}_l}, r_{\text{high}_l}] \\
\text{width}_l &= r_{\text{high}_l} - r_{\text{low}_l} = s_l
\end{aligned}$$

For a band $[u, v]$ inside a YES receipt's path, the YES cost is:

$$\text{YES_cost}[u, v] = \text{integral from } u \text{ to } v \text{ of } P_{\text{yes}}(r) \, dr$$

For a NO receipt that traversed the same band in the opposite direction, the NO cost is:

$$\text{NO_cost}[u, v] = \text{integral from } u \text{ to } v \text{ of } P_{\text{no}}(r) \, dr$$

Because binary LMSR marginal prices are complements:

$$\begin{aligned}
& \text{YES_cost}[u,v] + \text{NO_cost}[u,v] \\
&= \text{integral from } u \text{ to } v \text{ of } (P_{\text{yes}}(r) + P_{\text{no}}(r)) \, dr \\
&= \text{integral from } u \text{ to } v \text{ of } 1 \, dr \\
&= v - u
\end{aligned}$$

The band width $v - u$ is also the number of matched fixed-payout shares represented by that segment. So a matched YES segment and NO segment over the same LMSR path band contain exactly the collateral needed to mint one YES and one NO complete set per retained share.

Two receipts are compatible only over the positive-width intersection of their path intervals:

$$\begin{aligned}
\text{overlap_low} &= \max(r_{\text{low_yes}}, r_{\text{low_no}}) \\
\text{overlap_high} &= \min(r_{\text{high_yes}}, r_{\text{high_no}}) \\
\text{overlap_width} &= \text{overlap_high} - \text{overlap_low}
\end{aligned}$$

If:

$$\text{overlap_width} \leq 0$$

then the receipts do not match, even if aggregate YES shares, aggregate NO shares, or aggregate collateral appear balanced.

The filled market cap is the total matched path width after clearing:

$$F = \text{sum over cleared bands of matched_band_shares}$$

A market graduates when:

$$F \geq \text{graduation}_{\text{threshold}}$$

8. Graduation Auction And Path-Overlap Clearing Rule

Graduation clearing is a deterministic sweep over receipt path segments.

The clearing input is the frozen active receipt book:

$$\begin{aligned}
Y_{\text{lots}} &= \text{active YES receipt lots} \\
N_{\text{lots}} &= \text{active NO receipt lots}
\end{aligned}$$

Each lot starts with:

$$\begin{aligned}
\text{retained_shares}_1 &= 0 \\
\text{retained_cost}_1 &= 0 \\
\text{refund}_1 &= 0
\end{aligned}$$

The deterministic v0.2 clearing rule:

1. Convert every receipt lot into a path interval $[r_{\text{low}_1}, r_{\text{high}_1}]$.
2. Collect every r_{low_1} and r_{high_1} endpoint.
3. Sort and deduplicate endpoints.

4. Sweep each adjacent band $[r_k, r_{k+1}]$.
5. Ignore zero-width or dust-width bands.
6. Find YES lots whose path covers the whole band.
7. Find NO lots whose path covers the whole band.
8. If either active set is empty, the band has no match.
9. Let $m_k = \min(\text{active_yes_count}, \text{active_no_count})$.
10. Retain the scarce side fully in the band.
11. Retain the crowded side pro-rata within the band.
12. Add the retained band shares and retained band cost to each active receipt.
13. After all bands are swept, refund every receipt's unretained path cost.

For a band with width:

$$w_k = r_{k+1} - r_k$$

the retained fractions are:

$$\begin{aligned} \text{yes_fraction_k} &= m_k / \text{active_yes_count} \\ \text{no_fraction_k} &= m_k / \text{active_no_count} \end{aligned}$$

So each active YES receipt in the band retains:

$$\begin{aligned} \text{retained_yes_shares} &+= w_k * \text{yes_fraction_k} \\ \text{retained_yes_cost} &+= \text{YES_cost}[r_k, r_{k+1}] * \text{yes_fraction_k} \end{aligned}$$

and each active NO receipt in the band retains:

$$\begin{aligned} \text{retained_no_shares} &+= w_k * \text{no_fraction_k} \\ \text{retained_no_cost} &+= \text{NO_cost}[r_k, r_{k+1}] * \text{no_fraction_k} \end{aligned}$$

This is pro-rata, but only inside a specific path band. The protocol does not apply one global pro-rata haircut to all YES receipts or all NO receipts.

The matched market cap contributed by the band is:

$$F_k = w_k * m_k$$

and the locked collateral contributed by the band is:

$$\begin{aligned} L_k &= m_k * (\text{YES_cost}[r_k, r_{k+1}] + \text{NO_cost}[r_k, r_{k+1}]) \\ L_k &= m_k * w_k \\ L_k &= F_k \end{aligned}$$

This is the core solvency reason for path-overlap clearing: every retained band is backed exactly by the two opposite LMSR path segments that traversed that band.

Why Average-Price Matching Is Insufficient

A receipt's average entry price is lossless only if the entire receipt is retained or refunded. It is not lossless for partial fills. A YES receipt that moved the curve from 50% to 60% paid lower marginal prices near 50% and higher marginal prices near 60%. If only the 55% to 60% portion is matched by later NO demand, the retained portion has the cost of that upper path segment, not the whole receipt's average price.

This is why the protocol cannot just scale down the overbet side in aggregate. Different bettors received different path segments at different marginal odds. Full backing is proven band by band, where YES and NO marginal prices sum to one.

Partial Fills And Ties

Lots can be partially retained by band. A receipt can be fully retained over one path segment, pro-rata retained over another segment, and fully refunded elsewhere.

Within an exact same band, equal active segments receive equal retained fractions. Receipt sequence is still recorded for deterministic indexing, audit trails, and any future rounding tie-breaks, but normal v0.2 path-overlap clearing does not prefer earlier receipts over later receipts inside the same band.

Graduation Decision

The clearing engine computes:

$$F = \text{sum over bands of } F_k$$

If:

$$F \geq \text{graduation}_{\text{threshold}}$$

the market enters `Graduated`, retained receipts receive backed outcome tokens, and unretained exposure is refunded.

If clearing is run only because the pre-grad close time arrived and:

$$F < \text{graduation_threshold}$$

the market enters `NotGraduated` and all active receipts are refunded. In v0.2, sub-threshold matched exposure is not retained, because the graduation threshold defines the minimum viable market.

9. Refunds And Solvency

For each receipt lot:

$$\begin{aligned} \text{retained_shares_l} &= \text{sum retained band widths allocated to lot l} \\ \text{retained_cost_l} &= \text{sum retained band costs allocated to lot l} \\ \text{refund_l} &= \text{cost_basis_l} - \text{retained_cost_l} \end{aligned}$$

The refunded share count is:

$$\text{refunded_shares_l} = \text{provisional_shares_l} - \text{retained_shares_l}$$

But the refund amount is not generally:

$$\text{refunded_shares_l} * \text{average_entry_price_l}$$

That shortcut would reintroduce the same mixed-odds problem. Refunds are computed from the actual LMSR path cost that was not retained.

The user-level invariants:

$$\begin{aligned}
 0 &\leq \text{refund}_1 \leq \text{cost_basis}_1 \\
 \text{retained_cost}_1 + \text{refund}_1 &= \text{cost_basis}_1 \\
 \text{retained_cost}_1 &\leq \text{retained_shares}_1
 \end{aligned}$$

Solvency Theorem

Assume no trading fees. Let:

$$\begin{aligned}
 E &= \text{total escrowed receipt collateral} \\
 R &= \text{total refunds} \\
 L &= \text{total collateral locked for retained outcome tokens} \\
 F &= \text{total retained fixed-payout market cap}
 \end{aligned}$$

For every matched band k :

$$\begin{aligned}
 w_k &= \text{band width} \\
 m_k &= \text{matched segment count in the band} \\
 F_k &= w_k * m_k \\
 \text{YES_cost}_k + \text{NO_cost}_k &= w_k
 \end{aligned}$$

The collateral locked for that band is:

$$\begin{aligned}
 L_k &= m_k * (\text{YES_cost}_k + \text{NO_cost}_k) \\
 L_k &= m_k * w_k \\
 L_k &= F_k
 \end{aligned}$$

Summing across all matched bands:

$$\begin{aligned}
 L &= \sum L_k \\
 L &= \sum F_k \\
 L &= F
 \end{aligned}$$

All collateral not locked is refunded:

$$R = E - L$$

Therefore:

$$\begin{aligned}
 E &= R + L \\
 E &= R + F
 \end{aligned}$$

Since fixed-payout binary CTF-style settlement requires one unit of collateral per retained YES/NO complete set, and the clearing rule locks exactly F , the retained market is fully collateralized.

For any non-cancel YES/NO resolution:

$$\begin{aligned}
 \text{maximum_winner_payout} &\leq F \\
 \text{locked_collateral} &= F
 \end{aligned}$$

So the protocol never needs external subsidy, creator-bond collateral, insurance funds, or future fee revenue to pay retained outcome-token claims.

If fees are later introduced, they must appear explicitly in the conservation equation:

$$E = R + L + \text{fees}$$

and the protocol must specify whether fees are charged before clearing, from refunds, or from retained path cost. The v0.2 mechanism omits fees so the solvency accounting is transparent.

The market-level invariant for v0.2 is:

$$\text{total}_{\text{collateral locked}} = F$$

where F is the retained fixed-payout market cap.

10. Worked Example

Suppose $b = 100$. For readability, this example labels path bands by displayed probability, though the implementation clears on the deterministic coordinate $r = q_{\text{yes}} - q_{\text{no}}$.

Alice: YES moves the market from 50% to 60%
 Bea: YES moves the market from 60% to 70%
 Noah: NO moves the market from 70% to 65%

The NO receipt overlaps only the upper part of Bea's YES path:

Alice path: 50% to 60% -> no overlap with Noah
 Bea path: 60% to 70% -> overlap from 65% to 70%
 Noah path: 70% to 65% -> overlap from 65% to 70%

The matched width is:

$$\begin{aligned} w &= r(70\%) - r(65\%) \\ w &= 100 * (\ln(0.70 / 0.30) - \ln(0.65 / 0.35)) \\ w &\approx 22.8259 \text{ shares} \end{aligned}$$

Retained exposure:

Alice retains 0 YES
 Bea retains 22.8259 YES from the 65% to 70% band
 Noah retains 22.8259 NO from the 65% to 70% band

Refunds:

Alice receives her full receipt cost basis back.
 Bea receives back the actual LMSR cost of her unmatched 60% to 65% segment.
 Noah receives no unmatched-path refund because his whole path overlapped Bea.

Collateral locked:

$$\begin{aligned} \text{YES_cost}[65\%, 70\%] + \text{NO_cost}[65\%, 70\%] &= 22.8259 \\ \text{total locked} &= 22.8259 \end{aligned}$$

If YES resolves true:

Bea receives 22.8259
Noah receives 0

If NO resolves true:

Bea receives 0
Noah receives 22.8259

Alice's YES receipt did not fill because no opposite-side receipt traversed her 50% to 60% path band. The lower half of Bea's receipt did not fill because Noah's NO receipt did not traverse that band. Nobody loses unmatched exposure; unmatched path cost is refunded.

If the graduation threshold were higher than 22.8259, the same receipt book would not graduate in v0.2. All three bettors would be refunded because sub-threshold matched exposure is below the minimum viable market size.

11. Post-Graduation Interface

Post-graduation venue design is intentionally outside the core scope of this paper. After graduation, retained exposure can trade as CTF-style fixed-payout outcome tokens in any venue that preserves the complete-set invariant.

The graduation output is:

retained YES balances
retained NO balances
locked collateral backing complete sets
metadata and resolution rules

A mature venue can be a Polymarket-style central limit order book, a CTF-compatible automated market maker, a hybrid design, or another venue with equivalent settlement guarantees. The important design point is that the venue starts from backed exposure. It is not responsible for rescuing undercollateralized pre-grad receipts.

12. Comparison To Alternative Designs

Mechanism	Strength	Weakness	Why Not Default
Subsidized LMSR	Instant fills and clean probabilities	Requires protocol bankroll and can be farmed by informed flow	Violates no-subsidy goal
Pure pari-mutuel	Always solvent	Winners share a pot; final odds are not fixed at entry	Not the desired CTF-style payoff
Order book before graduation	Precise limit matching	Sparse early liquidity, complex UX	Feels slow for new markets
Constant-product automated market maker	Familiar automated market maker form	Liquidity providers take outcome risk	Needs liquidity provider capital before demand is proven
LMSR-priced path-overlap clearing	Smooth provisional pricing and fixed-payout settlement	No guaranteed fill, refunds can surprise users	Chosen v0.2 tradeoff

13. Settlement And Resolution

Trading mechanics determine which claims are retained. Resolution determines which retained claims win.

Resolution should be optimistic:

1. An artificial intelligence resolver proposes YES, NO, CANCEL, or another allowed result.
2. The proposal includes an evidence bundle.
3. A challenge window opens.
4. If nobody disputes, the proposal finalizes.
5. If disputed, the case escalates to human review.
6. High-value, conflicted, or appealed cases escalate to a final arbitration layer such as Universal Market Access or a similar optimistic oracle backstop.

The resolution layer should only operate on retained backed exposure:

$$\text{economic}_{\text{exposure}} = F$$

Removed and refunded exposure has no resolution rights.

14. Market Rule Schema

Every market should have a canonical rule spec:

Market rule schema

```
{
  "question": "Will X happen by DATE?",
  "outcomes": ["YES", "NO"],
  "yes_criteria": "Exact condition for YES.",
  "no_criteria": "Exact condition for NO.",
  "end_time": "2026-12-31T23:59:59Z",
  "resolution_deadline": "2027-01-02T23:59:59Z",
  "primary_sources": [],
  "fallback_sources": [],
  "source_priority_rule": "Use the first source in priority order that publishes a",
  "too_early_rule": "A proposal before the required source publishes is invalid.",
  "cancellation_rule": "Cancel only under explicit invalid-market conditions.",
  "evidence_requirements": ["url", "retrieval_timestamp", "claim_summary", "archive"]
}
```

The rule spec exists to prevent the market maker from becoming the court. Trading, clearing, and resolution are separate concerns.

15. Properties

No Subsidy

The mechanism never requires a protocol-funded market-maker loss budget. Incompatible exposure is refunded instead of subsidized.

Continuous Price Discovery

LMSR gives every pre-grad market an immediately visible price path. Even before final settlement liquidity exists, the market can communicate crowd belief.

Fixed-Payout Settlement

The retained post-graduation position is a YES or NO claim with a fixed terminal payout. It is not merely a pro-rata claim on a shared pot.

Deterministic Lifecycle

Every active receipt ends in one of two economic outcomes: retained backed exposure or refund. Markets that do not reach the graduation threshold do not create resolution-bearing claims.

Solvency By Construction

Only path-compatible matched exposure survives graduation. Since each retained YES/NO band locks exactly one unit of collateral per complete set, final claims can be backed.

Refundable Overbet Demand

Excess one-sided demand does not create bad debt. It creates refunds.

Resolution Modularity

The resolution layer can improve over time without changing the pre-grad clearing math.

16. Risks

Fill Misunderstanding

Users may interpret provisional receipts as guaranteed positions. The user interface must label them as provisional and show estimated retained shares and refunds.

One-Sided Markets

Very popular one-sided markets can create large refund events. This is economically safe but can disappoint users.

Path-Incompatible Demand

A market can have large nominal YES and NO demand but little path-compatible demand if the two sides traversed different LMSR path bands. Graduation should depend on path-compatible filled market cap, not headline provisional volume or aggregate collateral.

Clearing Rule Complexity

Batch clearing must be simple, deterministic, and heavily tested.

Locked Receipt UX

Non-transferable, non-withdrawable receipts make the protocol easier to reason about, but they also make pre-graduation participation less flexible than normal trading. Future versions can add receipt exits only if they preserve deterministic clearing and prevent low-cost price manipulation.

Manipulated Graduation

Attackers may attempt to create just enough path-compatible opposite-side demand to graduate a market on favorable terms. Graduation thresholds should include unique bettor counts, concentration checks, and resolution-quality gates.

Resolution Failure

If the oracle layer proposes the wrong result or resolves too early, retained positions can settle incorrectly. This is mitigated by evidence bundles, liveness windows, bonded disputes, human review, and final arbitration.

Ambiguous Markets

No clearing mechanism fixes ambiguous market wording. Market review and source-bound rules are required before pre-grad betting begins.

17. Conclusion

This paper specifies a no-subsidy prediction market launchpad that separates demand formation from final fill. LMSR provides continuous pre-graduation pricing. Path-overlap batch clearing determines how much exposure can safely become final. Incompatible and overbet path segments are refunded, so the protocol never needs to invent missing collateral.

The result is a market that can launch quickly, show a live probability curve, and remain solvent without a market-maker subsidy. The cost is explicit: pre-graduation bets are provisional. Fixed-payout exposure is earned at graduation by the presence of path-compatible opposing demand.

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